

HOW TO USE THIS ENCYCLOPEDIA

THE FOLLOWING PAGES WILL HELP YOU get the most out of your copy of the *Dorling Kindersley Illustrated Family Encyclopedia*. The encyclopedia consists of three volumes. Volumes 1–2 contain nearly 700 main entries organized alphabetically, from Aboriginal Australians through to Zoos. To find the entry you want, simply turn to the correct letter of the alphabet.

MEASUREMENTS AND ABBREVIATIONS

Most measurements are supplied in both metric and imperial units. Some of the most common abbreviations used in the encyclopedia are shown below in **bold type**.

°C = degrees Celsius
°F = degrees Fahrenheit
K = degrees kelvin
mm = millimetre; cm = centimetre
m = metre; km = kilometre
in = inch; ft = foot; yd = yard
g = gram; kg = kilogram
oz = ounce; lb = pound
ml = millilitre; l = litre
pt = pint; gal = gallon
sq km (km²) = square kilometre
sq ft (ft²) = square foot
kmh = kilometres per hour
mph = miles per hour
mya = million years ago
BC = before Christ
AD = anno Domini (refers to any date after the birth of Christ)
c. = circa (about)
b. = born; d. = died; r. = reigned

THE PAGE LAYOUT

The pages in this encyclopedia have been carefully planned to make each subject as accessible as possible. Main entries are broken down into a hierarchy of information – from a general introduction to more specific individual topics.

Alphabet locators

Letter flashes help you find your way quickly around the encyclopedia.

Sub-entries

Sub-entries provide important additional information and expand on points made in the introduction.

This sub-entry explains how rainbows are caused by raindrops in the air.

Diagrams

Clear diagrams help explain complex processes and scientific concepts.

The diagram here shows how a raindrop splits sunlight into its constituent colours.

If you cannot find the topic you want, then turn to Volume 3. This volume includes an index and gazetteer for the whole encyclopedia, which will direct you straight to the page you need. In addition, Volume 3 contains hundreds of reference charts, fact boxes, lists, and tables to supplement the information provided on the main entry pages.

Introduction

Clear introductions are the starting point for each entry. The introduction defines and provides an overview of each subject.

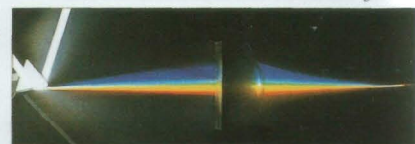
In the main entry on COLOUR, the introduction explains that colours are different forms of light, and that sunlight contains light of many different colours.

COLOUR

A WORLD WITHOUT COLOUR would be dull and uninspiring. Colour is a form of light. Light is made up of electromagnetic waves of varying lengths. The human eye detects these different wavelengths and sees them as different colours. White light – like that from the Sun – is a mixture of all the different wavelengths. Objects look coloured because they give out or reflect only certain wavelengths of light.

White light spectrum

Passing white light through a transparent triangular block called a prism separates out the different wavelengths of light. The prism refracts (bends) each wavelength by a different amount, forming a band of colours called a white light spectrum, or a visible spectrum. The seven main colours are red, orange, yellow, green, blue, indigo, and violet. Red has the longest wavelength and violet the shortest. Here, a convex lens combines the colours back into white light.



COLLEGES see SCHOOLS AND COLLEGES • COLOMBIA see SOUTH AMERICA

Rainbow

If it rains on a sunny day, you may well see a rainbow if you stand with your back to the Sun. A rainbow is a curved white light spectrum that forms when light is reflected and refracted by raindrops in the sky.



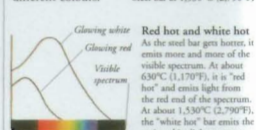
How a rainbow forms

When white sunlight passes through a raindrop, the raindrop acts like a tiny prism. The raindrop refracts the light and splits it up into its separate colours. The colours fan out and emerge as a spectrum. A rainbow is made up of spectra from millions of raindrops.



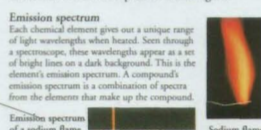
Colour and temperature

Objects at room temperature emit (give out) electromagnetic waves, but these waves are too long for human eyes to see. Heating an object, such as this steel bar, gives the waves it emits more energy and makes them shorter. The waves eventually become short enough to be seen, and the bar begins to glow. As the bar's temperature rises, it glows with different colours.



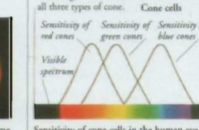
Spectroscope

An instrument called a spectroscope is used to analyze the light given out by hot substances. Inside the spectroscope, a prism or diffraction grating (a glass slide scored with fine lines) splits light from a glowing substance into its component wavelengths.



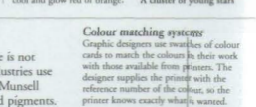
Cone cells

At the back of the eye there are special cells called cones that enable humans to see colours. There are three types of cone, called red, green, and blue cones. Each type of cone is sensitive to a different range of light wavelengths. White light stimulates all three types of cone.



Hot stars

The colour of a star gives a clue to its age. To the naked eye, most stars look white, but their true colours can be seen using a telescope. Young stars are hot and glow with white light. Older stars are relatively cool and glow red or orange.



Joseph von Fraunhofer

The German physicist Joseph von Fraunhofer (1787–1826) became interested in the nature of light while training as a mirror maker and lens polisher. His training enabled him to make spectroscopes of great precision. From 1814–17, he used them to make the first scientific study of the Sun's emission spectrum.

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Munsell colour system

Describing colours exactly using words alone is not easy. To avoid confusion, manufacturing industries use standard colour-identification systems. The Munsell system is used to specify colours for dyes and pigments. It defines a colour by its value (brightness), its chroma (strength), and its hue (position in the spectrum).



Colour matching systems

Graphic designers use swatches of colour cards to match the colours in their work with those available from printers. The designer supplies the printer with the reference number of the colour, so the printer knows exactly what's wanted.



Labels help to identify images.

act as rudders.

Strong chest muscles pull down the wings.

Penguin rises through the water to break through the surface.

Huddling reduces heat loss.

emperor penguins carry chicks around on their feet.

KING PENGUIN

SCIENTIFIC NAME *Aptenodytes patagonica*

ORDER Sphenisciformes

FAMILY Spheniscidae

DISTRIBUTION Islands and ocean north of Antarctica

HABITAT Coasts and open sea

DIET Fish and squid

SIZE Length, including tail 95 cm (37.5 in)

LIFESPAN About 20 years

Natural history data boxes

On the natural history pages, data boxes summarize essential information about a key animal featured in the entry. The box contains information about the animal's size, diet, habitat, lifespan, distribution, and scientific name.

This data box gives you key facts about the King Penguin.

Biography boxes

Most main entry pages have biography boxes that tell you about key people who have contributed to our knowledge of the subject. The encyclopedia also has single-page entries on the life and work of more than 50 major historical figures.

This biography box describes the work of the physicist Joseph von Fraunhofer.

Headings

The topic headings enable you to see at-a-glance which subjects are covered within the main entry.

The heading Colour matching systems refers to the way designers use reference numbers to match the colours on their work to the colours of printers' inks.